KING SALMAN INTERNATIONAL UNIVERSITY

Field Engineering Sciences

All Programs

Lecture : Data Resource Management

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By the end of this lecture, you should be able to...

Learning outcomes

- Explain why organizations need reliable, well-managed data.
- Distinguish characters, fields, records, files and databases.
- Explain the problems of separate file-processing systems.
- Describe the role of a Database Management System (DBMS).
- Recognize database types, data warehouses and data mining.
- Read a simple relational table and understand keys and relationships.
- Explain basic steps in planning and developing a database.

Why do we study data resource management?

Data is valuable only when it can be trusted and used.

Data supports decisions

- Engineers use measurements, test results and maintenance records.
- Organizations use data to plan work and monitor performance.
- Poor data can lead to poor decisions.

Data needs management

- Information must be accurate and up to date.
- Data should be organized and easy to retrieve.
- Access must be controlled and information protected.

What is data resource management?

This lecture follows the concepts in the original Data Resource Management course material.

Simple definition

Managing an organization’s data resources so people and systems can find and use the information they need.

What it involves

- Organizing data.
- Maintaining accuracy and consistency.
- Controlling access.
- Supporting retrieval, reporting and analysis.

From raw data to useful information

Reliable decisions depend on reliable data.

DATA

Individual facts or measurements.

Example: 24.8 °C

PROCESSING

Organizing, checking, combining or calculating data.

INFORMATION

A meaningful result.

Example: average temperature is 25.1 °C

DECISION

Use the information to choose an action, such as checking a cooling system.

Logical data elements: from small to large

These concepts form a useful data hierarchy.

CHARACTER

One symbol, such as A, 7 or #.

FIELD

One item of information, such as Student_ID or Temperature.

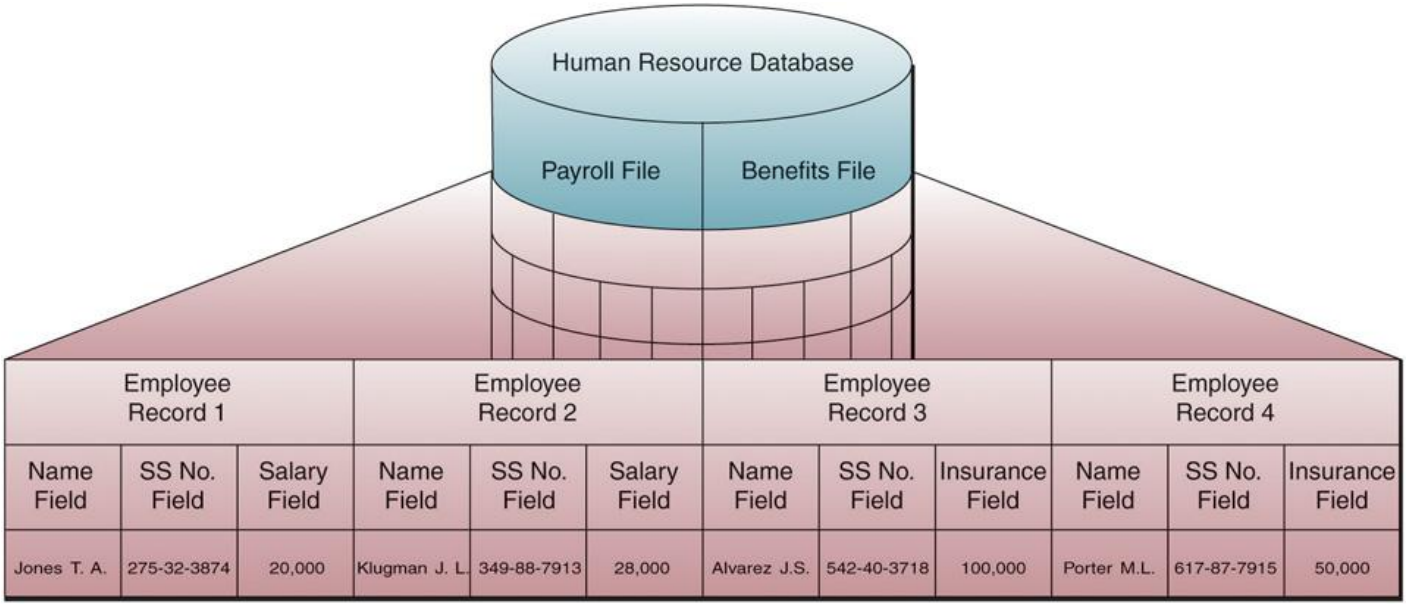
RECORD

A group of related fields describing one item or event.

FILE / DATABASE

A file groups related records; a database organizes related data for shared use.

Visual guide: logical data elements



Characters, fields, records and files form progressively larger data collections.

Example: an engineering measurement record

Good structure makes measurements easier to search and compare.

One record

Sensor_ID: S014
Location: Lab 2
Temperature: 24.8
Unit: °C
Timestamp: 10:30

Identify the elements

- Each named item is a field.
- The complete set describes one measurement event: a record.
- Many measurement records can be stored together.

Entities and attributes

An entity is the thing being described; attributes describe it.

Entity

A person, place, object or event about which data is stored.

Engineering examples:

- Sensor
- Machine
- Student
- Maintenance job

Attribute

A characteristic of an entity.

For a Sensor entity:

- Sensor_ID
- Type
- Location
- Calibration_Date

Relationships: how data connects

Relationships help avoid repeating the same facts in many places.

Example entities

STUDENT
Student_ID
Name

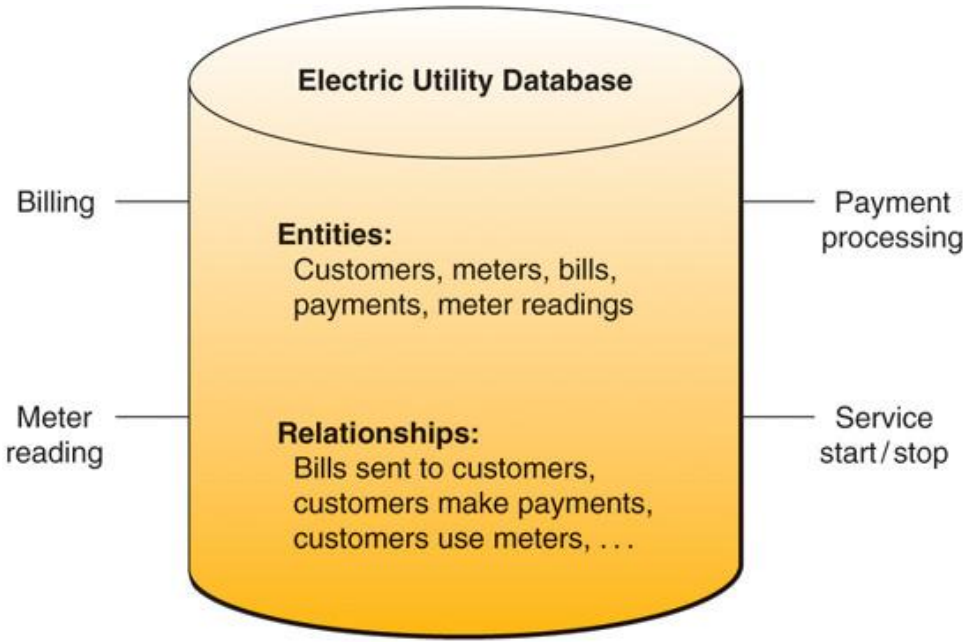
COURSE
Course_ID
Course_Name

A relationship

A student can enrol in several courses, and a course can include many students.

The database needs a way to represent this relationship.

Visual guide: entities and relationships



The example shows how entities and their data can be connected.

What is a database?

A database is more than a folder of unrelated files.

ORGANIZED

Data is arranged in a planned structure.

RELATED

Data elements connect to a common topic or process.

SHARED

Different users or applications can access appropriate data.

MAINTAINED

Data can be added, updated, searched and protected.

Traditional file processing

Separate files can work for small tasks, but sharing consistent data becomes harder as systems grow.

How it works

Different programs keep their own files.

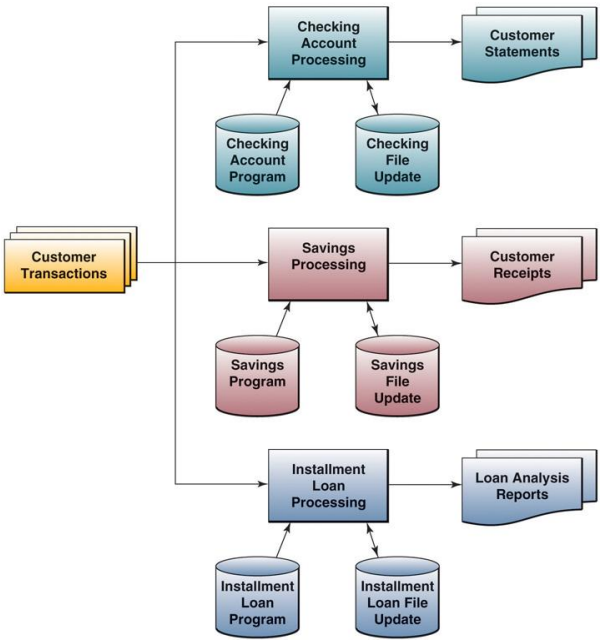
Example:

- Maintenance team has a machine file.
- Store team has a parts file.
- Production team has an equipment list.

Why it can be difficult

The same machine information may appear in several places, and the files may not agree.

Visual guide: traditional file processing



Separate applications maintain separate files, which can make shared information harder to manage.

Problems with separate files

These are key problems highlighted in the original course material.

REDUNDANCY

The same data is stored more than once.

INCONSISTENCY

Copies may show different values after updates.

LIMITED SHARING

Combining information from separate files can be difficult.

DATA DEPENDENCE

A change in file structure may require changes to programs.

A database management approach

A database approach can improve integration, but still needs good design and governance.

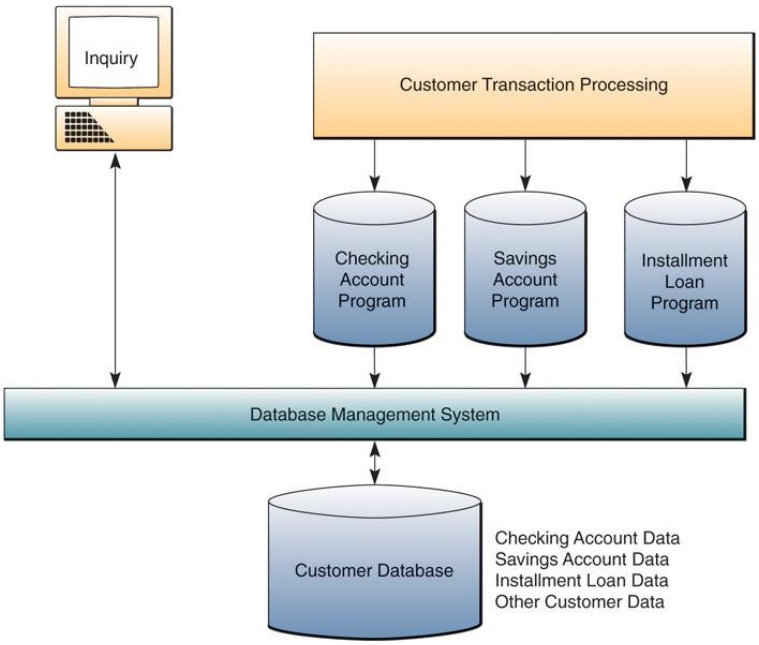
Shared data

Related data is organized in a common database that can serve several applications.

Benefits

- Less unnecessary duplication.
- More consistent updates.
- Easier sharing and reporting.
- Data structure can be managed separately from individual programs.

Visual guide: database management approach



A DBMS provides a shared interface between applications and the database.

What is a DBMS?

A DBMS is the software layer between users/applications and the database.

Database Management System

Software that helps create, maintain, query and control access to a database.

Common tasks

- Define tables and relationships.
- Add, update and delete records.
- Search and report data.
- Manage users and permissions.
- Support backup and recovery.

What can a DBMS help users do?

Different DBMS products provide different tools and features.

QUERY

Ask for selected data that meets conditions.

REPORT

Present results in a structured format.

MAINTAIN

Add new records and correct existing values.

DEVELOP

Support applications that use the database.

Database query vs. report

Queries retrieve selected data; reports present results for people to review.

Query

A request to retrieve data.

Example: show all machines whose last inspection date is more than 6 months ago.

Report

A formatted presentation of information.

Example: a monthly inspection report grouped by workshop and machine type.

A simple query: ask a clear question

Conceptual example; no database software is required

Question

Which sensors recorded temperatures above 30 °C?

Possible result

Sensor_ID | Temperature

S014 | 31.2 °C

S027 | 33.0 °C

Only matching records are returned.

Types of databases in the source material

The original material also introduces hypermedia databases.

Operational database

Stores detailed data needed for day-to-day operations.

Example: current machine status, work orders and inventory.

Distributed / external databases

Distributed: data is stored or replicated across connected sites.

External: data is obtained from outside sources or services.

Hypermedia databases

The central idea is that data items can be connected through links.

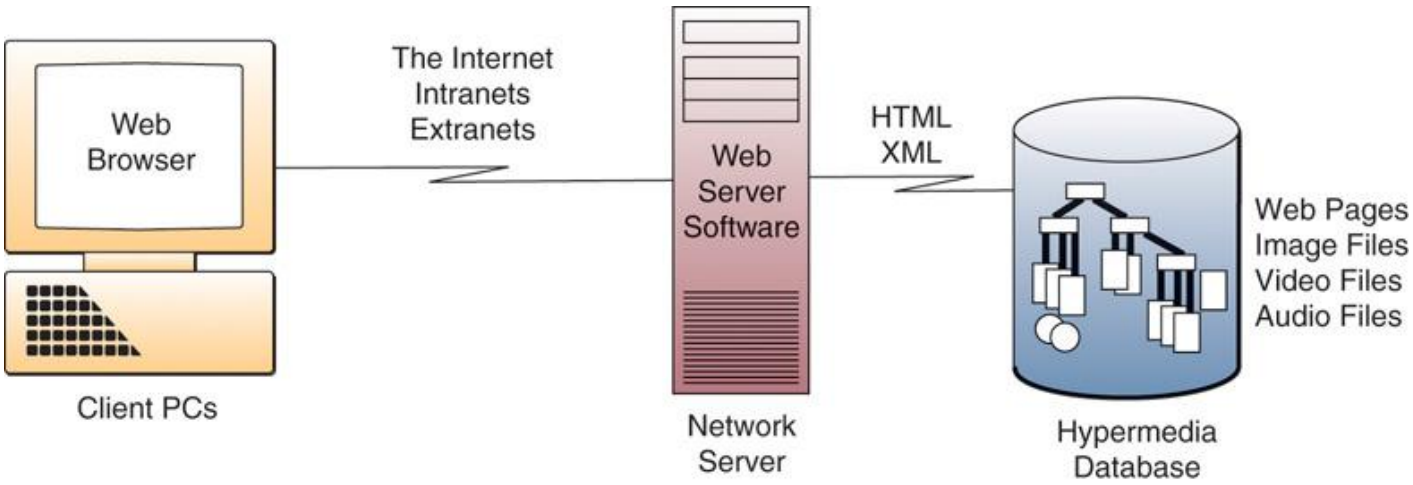
What they contain

Linked pages or items that may include text, images, audio, video and other media.

Example use

A digital maintenance guide links a machine model to diagrams, safety videos, parts lists and service instructions.

Visual guide: hypermedia database



Web technologies connect pages and multimedia through links.

Data warehouse

A warehouse supports analysis; it is not usually the main system for recording every new transaction.

Purpose

A large repository that brings data together from different operational and other sources for analysis.

Example

A university or engineering company combines several years of project, equipment, cost and performance data to examine long-term trends.

Data marts and data mining

Patterns can support investigation; they do not automatically prove cause and effect.

Data mart

A focused subset of data for a particular department or topic.

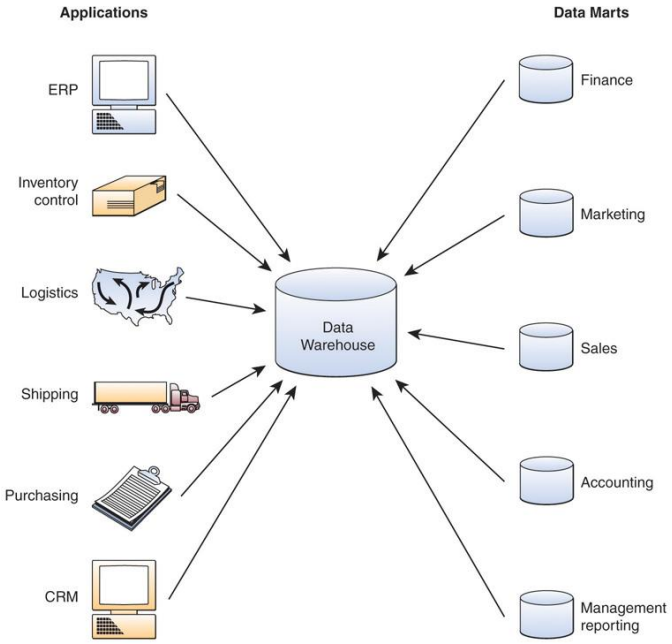
Example: a maintenance team’s view of machine failures and repair costs.

Data mining

Analyzing large collections of data to identify patterns or trends.

Example: finding which operating conditions are often associated with machine failures.

Visual guide: data warehouse and data marts



A central warehouse can supply focused data marts for different departments.

Data mining: engineering examples

Results depend on data quality, suitable methods and careful interpretation.

QUALITY

Find recurring patterns in product defects.

MAINTENANCE

Identify conditions often seen before equipment failure.

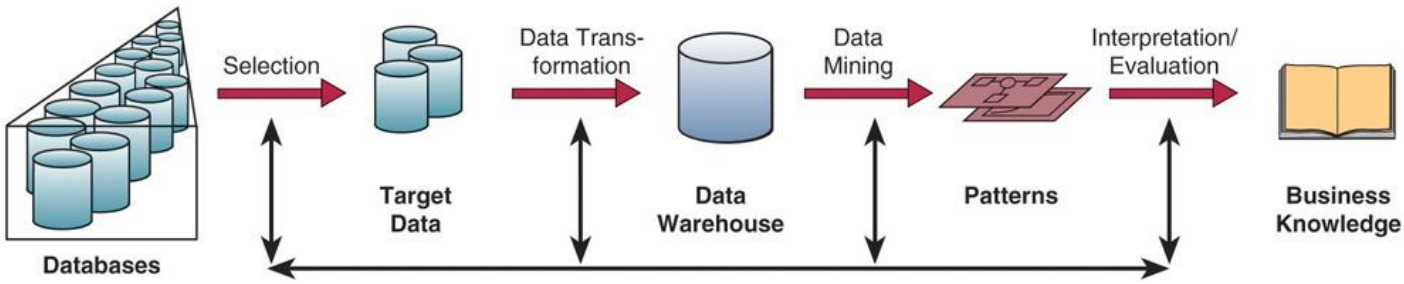
ENERGY

Compare consumption across buildings, times or operating conditions.

OPERATIONS

Explore delays, bottlenecks and repeated process problems.

Visual guide: data mining process



Data is selected and prepared before patterns are interpreted as useful knowledge.

Fundamental database structures

The original material also introduces multidimensional structures for analysis.

Relational

Data is organized in tables made of rows and columns.

Common in many business and engineering applications.

Other models from the source

Hierarchical: tree-like relationships.

Network: records can have multiple links.

Object-oriented: supports complex objects and associated operations.

Relational tables: rows and columns

A relational database may contain several linked tables.

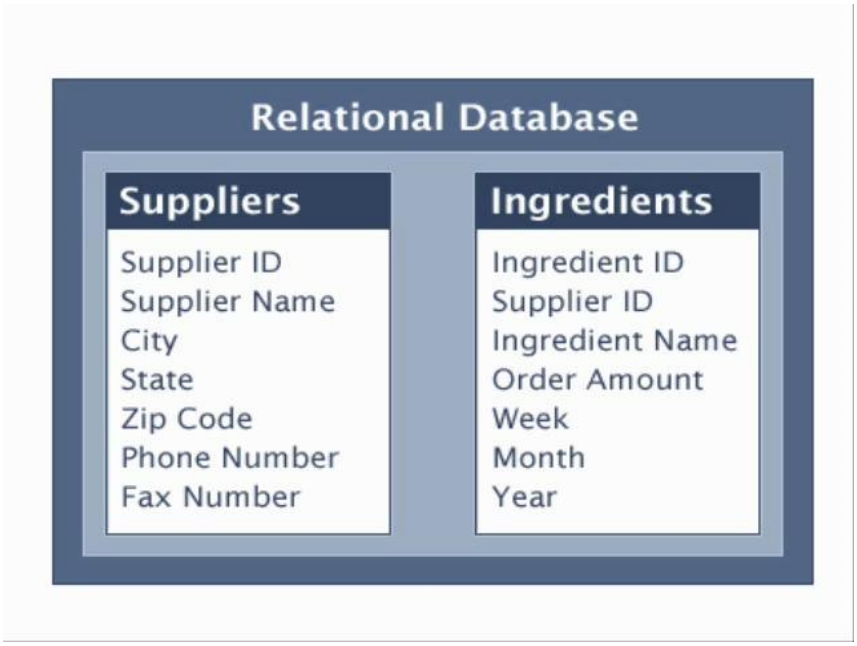
Example: SENSOR

Sensor_ID | Location | Unit
S014 | Lab 2 | °C
S027 | Workshop | °C
S031 | Lab 1 | %RH

Read the table

- Each row represents one sensor record.
- Each column represents an attribute.
- Sensor_ID identifies a sensor.

Visual guide: relational database



Relational data is represented in tables with fields and records.

Primary keys and foreign keys

Keys help link records while keeping data organized.

Primary key

A field (or set of fields) that uniquely identifies each record.

Example: Sensor_ID.

Foreign key

A field that refers to a key in another table to connect related records.

Example: a Measurement table stores Sensor_ID to show which sensor produced each reading.

Example: connect sensors and measurements

Sensor_ID links each measurement to the sensor that produced it.

SENSOR table

Sensor_ID | Location
S014 | Lab 2
S027 | Workshop

MEASUREMENT table

Measurement_ID | Sensor_ID | Value
M001 | S014 | 24.8
M002 | S014 | 25.1
M003 | S027 | 31.2

Why relationships matter

This reduces duplication and helps maintain consistent information.

Without a relationship

A measurement file may repeat the sensor location and type in every row. Updates can become inconsistent.

With a relationship

Store sensor details once in SENSOR and measurement values in MEASUREMENT. Use Sensor_ID to connect them.

Database development: start with the need

A good database starts with requirements, not with creating random tables.

Step 1 — understand the task

Ask who needs the data, what questions they need to answer and which processes use it.

Step 2 — identify the data

List entities, attributes, records and relationships.

Example: machines, inspections, technicians and spare parts.

A simple database development process

Database administrators and designers help manage these activities.

PLAN

Identify users, tasks and information needs.

MODEL

Define entities, attributes, keys and relationships.

BUILD & TEST

Create the structure, enter test data and check queries.

MAINTAIN

Correct data, manage access, back up and improve the system.

Data dictionary and metadata

Clear definitions help different users interpret data consistently.

Metadata

Data about data.

Example: a field called Temperature_Value has type decimal and unit °C.

Data dictionary

A catalog describing data fields, formats, meanings, allowed values and relationships.

Logical vs. physical views

Designers often separate the information model from the details of physical storage.

Logical view

What data is stored and how the data elements relate.

Example: Sensor, Measurement and their relationship.

Physical view

How the database is stored and accessed by the computer system.

Data quality: accuracy and consistency

A database cannot make poor input data trustworthy by itself.

Questions to ask

- Is the value correct?
- Is the unit recorded?
- Is the timestamp valid?
- Is a required field missing?
- Are duplicate records present?

Engineering example

A reading of 350 may mean 350 °C, 350 K or a sensor error if the unit and context are missing.

Data protection and responsible access

Reliable management includes availability, accuracy and appropriate access.

Protect the data

- Give users only the access they need.
- Keep backups.
- Use secure authentication.
- Follow organizational rules for sensitive data.

Use data responsibly

- Collect only what is needed.
- Correct errors when found.
- Avoid sharing private or confidential records without permission.

Worked example: maintenance database

Organized records can support maintenance planning and operational decisions.

Information needed

Machine_ID
Inspection_Date
Technician_ID
Fault_Code
Action_Taken
Downtime_Hours

Questions the database can answer

- Which machines have repeated faults?
- What is the average downtime?
- Which inspections are overdue?
- Which spare parts are frequently used?

Data resource management: original definition

Data Resource Management

Definition: A managerial activity that applies information systems technologies to the task of managing an organization's data resources to meet the information needs of their business stakeholders.

Why it matters

- Supports the information needs of users and stakeholders.
- Helps the organization manage data as a resource.
- Connects data management practices with business operations and decisions.

Logical data elements: original definitions

Basic elements

Character: a single alphabetic, numeric or other symbol.

Field: a group of related characters.

Entity: a person, place, object or event.

Attribute: a characteristic of an entity.

Larger elements

Record: a collection of attributes that describe an entity.

File: a group of related records.

Database: an integrated collection of logically related data elements.

Traditional file processing: full definition

Traditional file processing

Definition: Data are organized, stored and processed in independent files of data records.

In this approach, different applications or departments may maintain separate files.

Problems of file processing

• Data Redundancy: duplicate data requires an update to be made to all files storing that data.

• Lack of Data Integration: data stored in separate files require special programs for output, making ad hoc reporting difficult.

• Data Dependence: programs must include information about how data are stored, so a change in storage format requires a change in programs.

Database management approach: source definition

The database approach is more than placing files in one location.

Database management approach

Definition: Consolidates data records into one database that can be accessed by many different application programs.

- Provides a software interface between users and databases.
- Data definition is stored once, separately from application programs.

Database Management Software (DBMS)

Definition: Software that controls the creation, maintenance and use of databases.

The DBMS lets users and applications work with the database without each program independently managing the physical data files.

DBMS software components and uses

A DBMS provides tools for defining, storing, retrieving, maintaining and using data.

DBMS software components

• Database definition: language and graphical tools to define entities, relationships, integrity constraints and authorization rights.

• Nonprocedural access: language and graphical tools to access data without complicated coding.

• Application development: graphical tools to develop menus, data-entry forms and reports.

• Procedural language interface: combines nonprocedural access with full programming-language capabilities.

• Transaction processing: controls interference from simultaneous users and recovers lost data after failure.

• Database tuning: tools to monitor and improve database performance.

Uses of DBMS software

• Define and organize database structures.
• Store, update and retrieve data.
• Answer requests for information.
• Produce formatted reports.
• Maintain data accuracy as transactions and events occur.
• Support development of applications that use the database.

DBMS components and uses are both covered as separate source topics.

Visual guide: DBMS software components

Common DBMS Software Components	
• **Database definition**	Language and graphical tools to define entities, relationships, integrity constraints, and authorization rights
• **Nonprocedural access**	Language and graphical tools to access data without complicated coding
• **Application development**	Graphical tools to develop menus, data entry forms, and reports
• **Procedural language interface**	Language that combines nonprocedural access with full capabilities of a programming language
• **Transaction processing**	Control mechanisms to prevent interference from simultaneous users and recover lost data after a failure
• **Database tuning**	Tools to monitor and improve database performance

The source groups common DBMS tools by their functions.

Database interrogation, queries and reports

The source uses database interrogation for retrieving information in response to user requests.

Database interrogation

Definition: The capability of a DBMS to report information from the database in response to end users' requests.

Query Language: allows easy, immediate access to ad hoc data requests.

Report Generator: allows quick, easy specification of a report format for information users have requested.

Query vs. report; natural language vs. SQL

- Query: asks for selected data that meet conditions.
- Report: presents information in a specified, organized format.
- Natural-language query: a request phrased in everyday language.
- SQL query: a request written using Structured Query Language (SQL), a formal language for working with relational databases.

Database maintenance and application development

Databases must be updated, corrected and used by applications throughout their life.

Database maintenance

Definition: Updating a database continually to reflect new business transactions and other events.

It also includes updating a database to correct data and ensure the accuracy of the data.

Application development

The old source explains that end users, systems analysts and other application developers can use:

- Internal 4GL programming languages.
- Built-in software development tools provided by many DBMS packages.

These tools can develop custom application programs that work with the database.

4GL means fourth-generation language; this source term is retained.

Database types: original definitions

Operational and distributed databases

Operational database: stores detailed data needed to support the business processes and operations of a company.

Distributed database: databases that are replicated and distributed in whole or in part to network servers at a variety of sites.

External and hypermedia databases

External database: contains information available from commercial online services and many sources on the World Wide Web.

Hypermedia database: consists of hyperlinked pages of multimedia.

Data warehouse, data mart and data mining: definitions

Keep the distinctions between analysis storage, focused subsets and pattern discovery clear.

Data warehouse and data mart

Data warehouse: a large database that stores data extracted from various operational, external and other databases of an organization.

For analysis, the source summary describes warehouse data as cleaned, transformed and cataloged.

Data mart: a database that holds a subset of data from a data warehouse and focuses on a specific aspect of a company, such as a department or business process.

Data mining

Definition: Analyzing the data in a data warehouse to reveal hidden patterns and trends in historical business activity.

Patterns need careful interpretation before decisions are made.

Data mining uses

The original business examples are preserved alongside engineering examples.

Original uses

- Market-basket analysis: identify new product bundles.
- Find root causes of quality or manufacturing problems.
- Prevent customer attrition and acquire new customers.
- Cross-sell to existing customers.
- Profile customers with more accuracy.

Engineering links

- Find recurring patterns in product defects.
- Investigate conditions associated with equipment failure.
- Compare energy consumption across buildings or time periods.
- Explore repeated process delays and bottlenecks.

A pattern is not automatic proof of cause and effect.

Database structures: hierarchical, network and relational

Hierarchical and network structures

Hierarchical structure: relationships between records form a hierarchy or tree-like structure.

Network structure: data can be accessed by one of several paths because any data element or record can be related to any number of other data elements.

Relational structure

Definition: All data elements within the database are viewed as being stored in the form of simple tables.

Tables organize information in rows and columns. Keys can represent relationships between tables.

Model names and definitions are explicitly preserved.

Database structures: multidimensional and object-oriented

Multidimensional structure

Definition: A variation of the relational model that uses multidimensional structures to organize data and express the relationships between data.

It may support analysis across dimensions such as time, location or product.

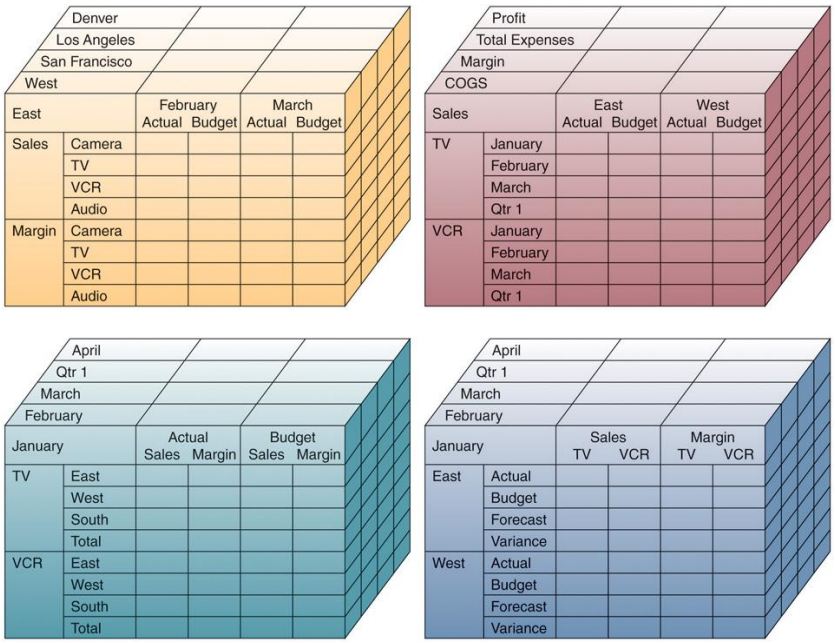
Object-oriented structure

Definition: Can accommodate more complex data types, including graphics, pictures, voice and text.

Encapsulation: data values and the operations that can be performed on them are stored as a unit.

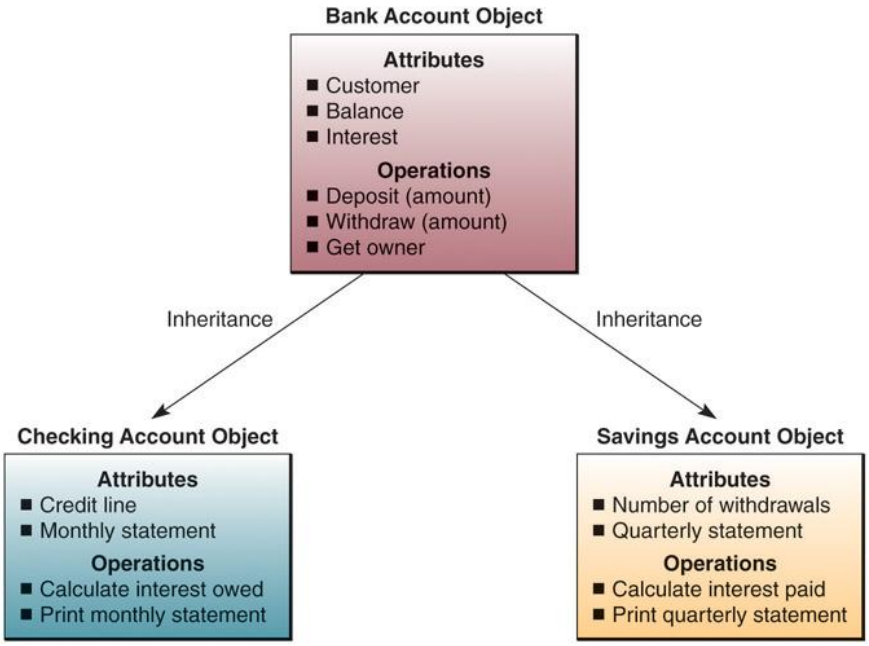
Inheritance: automatically creating new objects by replicating some or all characteristics of one or more existing objects.

Visual guide: multidimensional database



Multidimensional structures organize information across several dimensions.

Visual guide: inheritance in an object-oriented database



A specialized object can inherit attributes and operations from a more general object.

Evaluation of database structures

Source evaluation

- Hierarchical: best for structured, routine types of transaction processing.
- Network: best when many-to-many relationships are needed.
- Relational: best when ad hoc reporting is required.

How to use the comparison

- Consider the shape of the data and the relationships to represent.
- Consider how users need to retrieve and report information.
- No one model is automatically best for every application.
- The source's stated strengths are retained.

Database development: DBA, data dictionary and metadata

The source identifies responsibilities and definitions used during database development.

Database administrator (DBA)

Enterprise-wide database development is usually controlled by database administrators (DBAs).

The DBA helps coordinate the shared database structure and its development across an organization.

Data dictionary and metadata

Data dictionary: a catalog or directory containing metadata.

Metadata: data about data.

A data dictionary helps users and designers understand data definitions and descriptions.

DBA, data dictionary and metadata remain explicit terms.

Data planning, data modeling and ER diagrams

Planning and modeling define what the database should represent before it is built.

Data planning and data modeling

Data planning: database administrators and designers work with corporate and end-user management to develop an enterprise model that defines the basic business processes of the enterprise.

Data modeling: the process where relationships between data elements are identified.

Entity-relationship diagram (ERD)

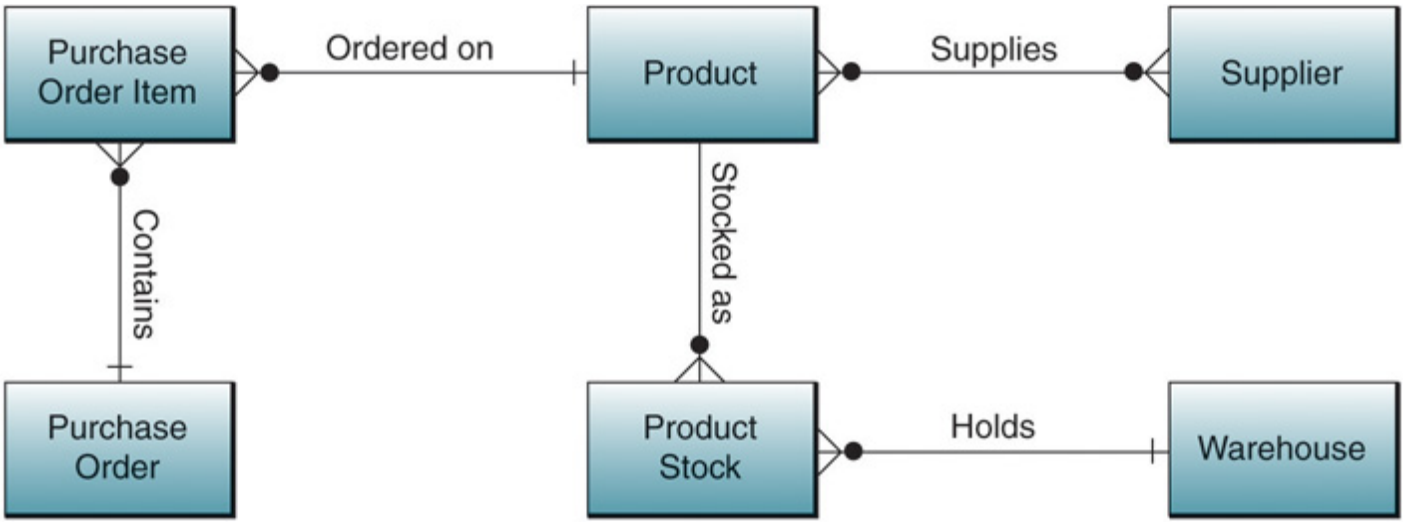
An entity-relationship diagram represents entities and the relationships between them.

It helps designers plan how data items connect before database structures are implemented.

Example: SENSOR — produces — MEASUREMENT.

ER diagram is included as a separate source topic.

Visual guide: entity-relationship diagram



The diagram shows entities and relationships before implementation.

Logical and physical database views: source definitions

The logical view describes meaning and relationships; the physical view describes storage and access.

Logical view

Definition from the source: data elements and relationships among them.

Example: which entities exist, which attributes describe them and how they relate.

Physical view

Definition from the source: describes how data are to be stored and accessed on the storage devices of a computer system.

Logical models and physical storage designs are related, but they describe different levels.

Hypermedia databases: web technologies in the source diagram

Web environment

- Internet: a global network connecting networks and devices.
- Intranet: a private network used within an organization.
- Extranet: controlled access to selected resources for external parties.
- Web browser: client software used to access web pages.
- Web server software: delivers web resources to clients.

Web content and hypermedia

- HTML: HyperText Markup Language, used to structure web pages.
- XML: eXtensible Markup Language, used to describe and structure data.
- Hypermedia database: linked pages containing text, graphics, images, audio, video and other media.
- Links connect related pages or media items.

Data warehouse system: source architecture

Sources and data management

- Operational, external and other databases provide source data.
- Data acquisition: capture, clean, transform, integrate and store data.
- Warehouse design defines how the warehouse is organized.
- Data management organizes and controls warehouse data.
- Metadata management manages descriptions of the data.

Warehouse outputs and analysis

- Enterprise data warehouse: a central store for integrated organizational data.
- Data marts: focused subsets for departments or processes.
- Analytical data store supports analysis.
- Data analysis tools, query/report tools and web information systems help users access and interpret results.

Data mining: process from data to knowledge

Process stages

1. Data sources provide the starting information.
2. Selection identifies target data for the task.
3. Data transformation prepares the selected data.
4. Data warehouse stores organized data for analysis.
5. Data mining searches for patterns.

Interpretation and use

6. Patterns are produced by the analysis.
7. Interpretation and evaluation assess whether the patterns are useful and reliable.
8. Business knowledge is developed from results that are understood in context.

Data quality and expert interpretation matter at every stage.

Database development process: all five source steps

Steps 1–3

1. Data Planning: develops a model of business processes.
2. Requirements Specification: defines information needs of end users in a business process.
3. Conceptual Design: expresses all information requirements as a high-level model.

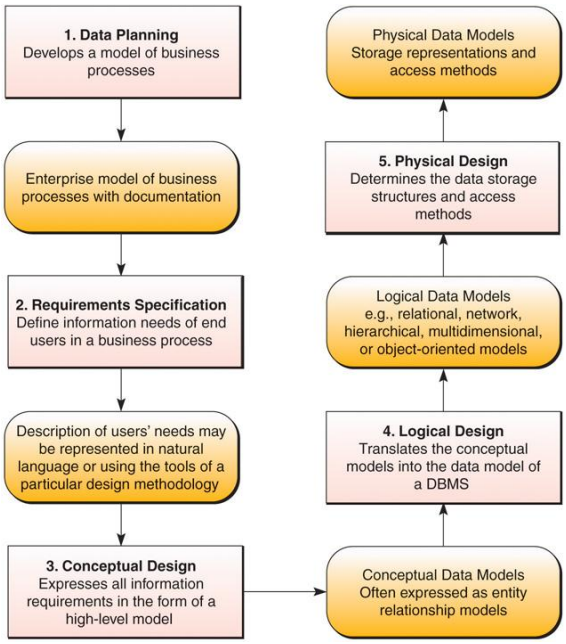
Users' needs may be described in natural language or through a design methodology.

Steps 4–5 and data models

4. Logical Design: translates conceptual models into the data model of a DBMS. Logical data models may be relational, network, hierarchical, multidimensional or object-oriented.
5. Physical Design: determines data storage structures and access methods.

Physical data models represent storage and access methods.

Visual guide: database development process



Planning, requirements, conceptual design, logical design and physical design are distinct steps.

Logical user views, schema and physical data views

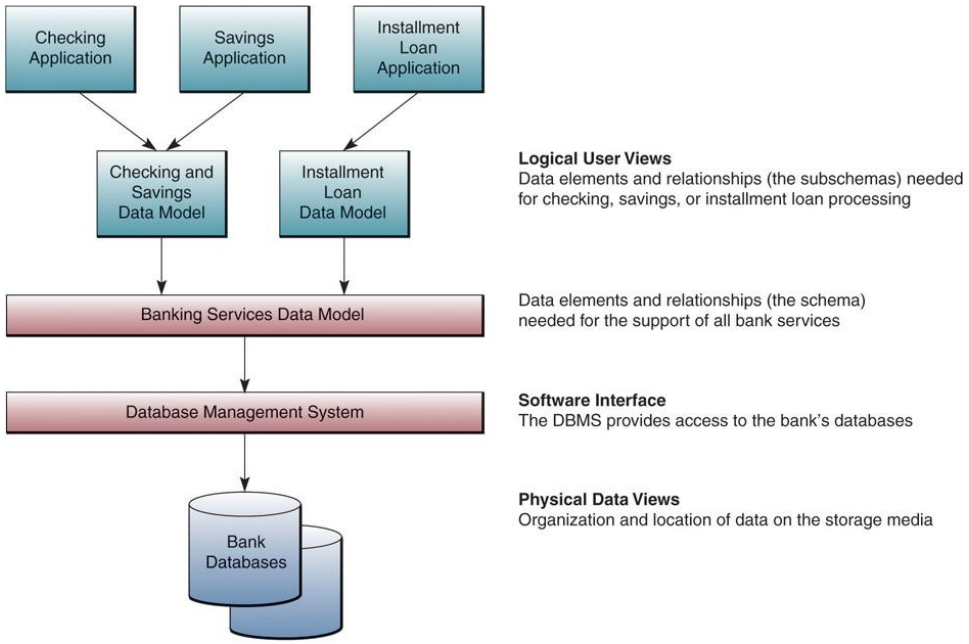
Logical views and schema

- Logical user views: data elements and relationships (subschemas) needed for a particular application, such as checking, savings or loan processing.
- Schema: data elements and relationships needed to support the full set of organizational services.
- Different applications may use different logical views of shared data.

Software interface and physical views

- Software interface: the DBMS provides access to the organization's databases.
- Physical data views: the organization and location of data on storage media.
- Application views, the shared schema and physical storage describe different levels of the same database system.

Visual guide: logical and physical views



Logical views describe data and relationships; physical views describe storage and access.

30-minute Take-home

Be ready to explain your choices to the class.

Activity A — model the data (15 min)

In small groups, design a simple database for a university engineering laboratory.

Identify 3 entities, 3–5 attributes for each, and at least one relationship.

Activity B — spot the file problem (15 min)

A machine's location is copied into four separate files. The machine moves, but only two files are updated.

Explain the problem and propose a database-based solution.

Quick practice: check your understanding

Answer individually, then compare with a partner.

Key points

- Put these in order from smallest to largest: database, character, file, field, record.
- Give one example of an entity and two attributes.
- Name two problems caused by keeping separate files.
- What does a DBMS do?
- Explain the difference between a data warehouse and a data mart.
- Why is a primary key useful?

Lecture 3 Review: Multiple Choice — 1.1

Choose the best answer. Check the answer key only after all questions.

Question 1

Which sequence correctly moves from a small data element to a larger collection?

A. Database → file → field → character

B. Character → field → record → file → database

C. Record → character → database → field

D. Field → character → file → record

Question 2

In a sensor database, what is Sensor_ID most likely to be?

A. A report format

B. A data warehouse

C. A field that identifies a sensor

D. A database application

Lecture 3 Review: Multiple Choice — 1.2

Choose the best answer. Check the answer key only after all questions.

Question 3

What is a major problem with storing the same machine location in several separate files?

A. It guarantees better accuracy

B. It removes all duplication

C. It makes every query automatic

D. Copies may become inconsistent

Question 4

What is the main role of a DBMS?

A. Manage the creation, use and maintenance of databases

B. Replace all input devices

C. Measure physical values directly

D. Design computer processors

Lecture 3 Review: Multiple Choice — 2.1

Choose the best answer. Check the answer key only after all questions.

Question 5

Which database structure organizes data in rows and columns?

A. Hierarchical only

B. Relational

C. Network only

D. Sequential file

Question 6

What does a primary key do?

A. Stores a backup copy

B. Formats a printed report

C. Uniquely identifies a record

D. Converts data into an image

Lecture 3 Review: Multiple Choice — 2.2

Choose the best answer. Check the answer key only after all questions.

Question 7

What is a data warehouse mainly used for?

A. Typing individual records only

B. Replacing all operational systems

C. Controlling a sensor directly

D. Bringing together data for analysis

Question 8

Which example best describes data mining?

A. Searching for patterns in machine failure records

B. Renaming a file

C. Adding a keyboard

D. Formatting a table heading

Lecture 3 Review: Multiple Choice — 3.1

Choose the best answer. Check the answer key only after all questions.

Question 9

What is metadata?

A. Duplicate data

B. Data about data

C. A type of input device

D. A computer processor

Question 10

Which is an example of a foreign key?

A. A report title

B. A unit label only

C. Sensor_ID in a measurement table that refers to the SENSOR table

D. A database file name

Lecture 3 Review: Multiple Choice — 3.2

Choose the best answer. Check the answer key only after all questions.

Question 11

What does a data dictionary describe?

A. Only the number of computers in a lab

B. How to print slides

C. The speed of a CPU

D. Data fields, meanings, formats and rules

Question 12

Why is data quality important in engineering?

A. Incorrect or unclear measurements can lead to poor decisions

B. It makes units unnecessary

C. It guarantees every prediction is correct

D. It removes the need for testing

Lecture 3 Review: Answer Key

Check your answers after completing all 12 questions.

Questions 1–4

1. B — Data hierarchy
2. C — Sensor identifier
3. D — Inconsistent copies
4. A — Manage databases

Questions 5–8

5. B — Relational
6. C — Uniquely identifies a record
7. D — Analysis across sources
8. A — Find patterns

Questions 9–12

9. B — Data about data
10. C — Links tables
11. D — Describes data definitions
12. A — Supports reliable decisions

Lecture summary

Remember the concepts and how they connect.

Key points

- Data resource management organizes data so people can use it reliably.
- Characters, fields, records, files and databases are related levels of data organization.
- Separate files can create duplication, inconsistency, limited sharing and data dependence.
- A DBMS supports database creation, maintenance, queries, reports and access control.
- Relational databases use tables connected by keys and relationships.
- Data warehouses support analysis; data mining looks for useful patterns.
- Database planning, modeling, testing, quality and protection are all important.

Exit ticket & next lecture

One-minute individual reflection

Before you leave

1. Explain one difference between a file and a database.
2. Give one example of a primary key.
3. Describe one problem a DBMS can help reduce.

Next lecture: Telecommunications & Networks

We will examine how computers and information systems communicate and share data.

Useful videos

- https://www.youtube.com/watch?v=PZVIV8wC4gM
- https://www.youtube.com/watch?v=5kiMg7GXAsY
- https://www.youtube.com/watch?v=U5Gg9R-Dc0Q
- https://www.youtube.com/watch?v=fcsXVJyxFbA

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